

DAYANANDA SAGAR COLLEGE OF ENGINEERING

(An Autonomous Institute affiliated to Visvesvaraya Technological University (VTU), Belagavi,
Approved by AICTE and UGC, Accredited by NAAC with 'A' grade & ISO 9001 – 2015 Certified Institution)
Shavige Malleshwara Hills, Kumaraswamy Layout, Bengaluru-560 111, India

DEPARTMENT OF ARTIFICIAL INTELLIGENCE AND MACHINE LEARNING



INTERNSHIP REPORT ON

“Developing Applications and Integration in ServiceNow”

Submitted in partial fulfillment for the award of the degree of

Bachelor of Engineering

In

Artificial Intelligence and Machine Learning

Submitted by:

Hemanth Gowda V - 1DS21AI022

Undertaken at



Royal Cyber Inc.

Unit No.07 & 08, 7th Floor, Innovator Building, ITPB,
Whitefield Road, Bangalore 560066

**Department of Artificial Intelligence and Machine Learning,
Dayananda Sagar College of Engineering, Bengaluru**

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CERTIFICATE

This is to certify that the Internship titled **“Developing Applications and Integration in ServiceNow”** carried out by **Hemanth Gowda V [1DS21AI022]**, a bonafide student of **Dayananda Sagar College of Engineering**, in partial fulfillment for the award of **Bachelor of Engineering in Artificial Intelligence and Machine Learning** during the academic year **2024–2025**. It is certified that all corrections/suggestions indicated have been incorporated in the report. The internship report has been approved as it satisfies the academic requirements in respect of the course Internship (21INT82).

Signature of the
Reporting Manager
Vellore Abdullah Basha,
Director of Business Operation,
Royal Cyber Inc

Signature of the
Internship Coordinator
Prof. Kavya D N,
Assistant Professor,
Dept. of AI & ML, DSCE

Signature of the
Head of the Department
Dr. Vindhya P Malagi,
Professor & Head,
Dept. of AIML, DSCE

Name of the Examiners

Signature with date

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2.

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DEPARTMENT OF ARTIFICIAL INTELLIGENCE AND MACHINE LEARNING



DECLARATION

I, **Hemanth Gowda V**, a final year student of **Artificial Intelligence and Machine Learning, Dayananda Sagar College of Engineering, Bengaluru**, hereby declare that this internship report is based on the work undertaken during my ongoing internship in the role of **ServiceNow Developer Intern**, at **Royal Cyber Inc, Bengaluru**.

This report is submitted in partial fulfillment of the requirements for the award of the Bachelor's Degree in Artificial Intelligence and Machine Learning during the academic year 2024–2025.

Date: 12/04/2025

Place: Bengaluru

Internship Agreement



Bangalore Address:-

Unit No.07 & 08, 7th Floor, Innovator Building, International Tech Park, Bangalore (ITPL), Whitefield Road, Bangalore – 560 066.

p : +91-80-4098 3128

Chennai Address:-

5th Floor RR TOWERS –
5 No : 33 A South Phase, Developer Flats, Thiru.Vi.ka Industrial Estate, Guindy, Chennai 600032

p : +91-44-4230 1500

CIN : U72200KA2013PTC069261

Email : info@royalcyber.com

Web : www.royalcyber.com

lnkd.in/fddGmAY

twitter.com/RoyalCyberUSA

facebook.com/RoyalCyber/

goo.gl/RGWk5W

THIS INTERNSHIP AGREEMENT made as 07th February 2025 between **Royal Cyber Private Limited**, a corporation incorporated under the laws of India, and having its offices at **Unit No.07 & 08, 7th Floor, Innovator Building, ITPB, Whitefield Road, Bangalore 560 066 (“Company”)** and **Hemanth Gowda V.** residing at C/o Venkatsh Bairapattna, Channapatna, Karnataka, Pincode- 562160 India. Pan Card No.-BUIPH2880J (the “Intern” or “you”).

Pursuant to your Internship application to the Company, we are pleased to inform you that we would like to engage you as an Intern in the department of Digital Transformation on the basis of the terms mentioned below. The purpose of this Internship Agreement is to enable the Intern to gain knowledge and experience during the course of this Internship Agreement.

1. Date of Commencement

Your engagement with Royal Cyber will commence on 10th **February 2025** and the same shall end upon the completion of the Internship period or upon termination of this Intern Agreement as per the terms agreed herein.

2. Working Hours:

The working hours shall be 2:00pm to 11:00pm IST which shall be subject to change from time to time as per the business requirement of the Company.

3. Term:

The initial term of this engagement shall be for a period of Six (6) months. However, the Internship may be extended by the Company, depending upon the intern’s performance and evaluation by the supervisor.

4. Compensation:

As remuneration, we are offering you a monthly stipend of INR 20,000/- (Twenty Thousand only) per month subject to deduction of applicable taxes or withholdings as per law, which shall be paid to you on the last working day of each month.

5. Internship:

Your engagement pursuant to this Internship Agreement does not constitute permanent/full time employment and you will not be entitled to any employment benefits that permanent/full time Interns of Royal Cyber are entitled to. In the event of there being a suitable project/assignment, which suits your expertise and capabilities, Royal Cyber may offer you to provide you with an opportunity with Part Time or full time offer of employment, upon acceptance of which, this Letter of Engagement will stand terminated. The detailed terms governing your employment then, will be as set out in the offer of employment.

DocuSigned by:

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Microsoft
GOLD CERTIFIED
Partner

IBM
Gold Partner

SAP
Gold Partner

salesforce
Crest
Partner

aws

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MuleSoft

Google Cloud databricks

6. Responsibility of Intern:

- a) Intern shall adhere to all the relevant and applicable policies of the Royal Cyber such as Code of Conduct Policy, Sexual Harassment Policy, Anti-Corruption Policy and any other Policy which may be applicable to the Interns from time to time.
- b) Intern shall execute Non-Disclosure Agreement and Intellectual property Rights Agreement which is annexed herein as **annexure A** of this Policy with the Royal Cyber wherein Intern shall ensure that the confidential Information, Trade Secrets, Customer Data, Intellectual Property Rights and other relevant data is protected and kept confidential.
- c) Intern shall ensure all the projects or Tasks allocated to the Intern is completed in timely manner.
- d) Intern shall maintain professional attitude and behavior during the term of the internship and Intern shall ensure that he/she performs their duties diligently with utmost care and caution.

7. Performance Evaluation:

- a) Interns shall be subjected to evaluation of their on their performance based on the goals and projects/tasks allocated and set at the beginning of the internship.
- b) The evaluation shall be done by the intern's supervisor and/or mentor.
- c) All the relevant feedback shall be shared with the intern from time to time.

8. A. Termination of Internship:

- a) Termination by Company: Company may terminate the Internship at any time, without any cause.
- b) Company reserves the right to terminate an internship at any time if the intern breaches any of the company's policies or guidelines or if the performance of the Intern is found to be unsatisfactory.

8. B. Termination by Intern:

- a) Intern may terminate the Internship by giving prior written seven (7) notice to Company.

9. No Employer and Intern Relationship:

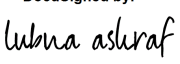
Internship offered by Company shall not create or constitute a relationship of employer and Intern. The internship model is solely designed to ensure that the intern shall learn and progress in their respective career.

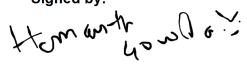
10. Grievance Redressal:

- a. In the event an Intern requires to raise any grievances or concerns which an Intern may have during the term of their internship, the said grievance shall be raised by the intern with the respective supervisor or Human Resource Manager or any person who may be appointed by Company in this behalf.
- b. The grievances shall be addressed in a timely and confidential manner by Company.

11. Governing Law and Jurisdiction:

The rules and regulations of this policy shall be governed by the **laws of India** and courts situated at **Bengaluru** shall have an exclusive jurisdiction to try any dispute in relation to this policy.

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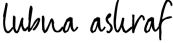
12. Company Policy:

An intern at Royal Cyber, you are expected to adhere to the company's code of conduct, policies, Sexual Harassment Policy, Anti Bribery Policy and any other Policy which may be applicable to you from time to time which shall be subject to change as per Company's discretion..An Intern shall maintain a professional and respectful attitude, maintain the privacy and security of company information, and complying with all relevant laws and regulations.

The Intern shall have the right to seek independent legal advice in terms of this Internship Agreement.

Please confirm your acceptance of this internship offer by signing below:

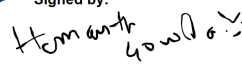
For Royal Cyber Pvt Ltd

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Lubna Ashraf

Global HR Head

I, **Hemanth Gowda**, agree that I have read, understood, and I voluntarily accept and sign this as per the terms & conditions stated above:

Signature: 
Signed by:
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Name: Hemanth Gowda

Date: 2/7/2025 | 9:54 AM PST

Annexure A

This Agreement is entered into by and between **Royal Cyber Private Limited**, having its office at Royal Cyber Pvt. Ltd. Unit No.07 & 08, 7th Floor, Innovator Building, International Tech Park, Bangalore (ITPL), Whitefield Road, Bangalore – 560 066., India. (**hereinafter called the “Royal Cyber” or “Company”**)

Hemanth Gowda V, an Indian citizen, aged **21** years, permanently residing at C/o Venkatsh Bairapattna, Channapatna, Karnataka, Pincode- 562160 India.. (**hereinafter called the “Intern”**)

The Intern will have access to non-public and confidential information relating to Company during the course of Employment with Royal Cyber and in consideration of the parties' discussions and any access You may have to non- public information of Royal Cyber, you hereby agrees as follows:

1. Definitions

In this Agreement, the following terms shall have the following meanings:

“Confidential Information” means Inventions and any other information or data of whatever nature (technical, financial, scientific, commercial or otherwise) which is disclosed to You by or on behalf of Royal Cyber, or which You otherwise become aware of during the course of Your employment, or which is generated by You in the course of your employment, and shall include but shall not be limited to trade secrets, or any information which is in the nature of a trade secret, information supplied to Royal Cyber under an obligation of confidence, and information about the business of Royal Cyber, its Interns, agents, vendors, subcontractors, partners and clients.

“Intellectual Property Rights” means patents, copyright, designs, trade or service marks, semiconductor topography rights, database rights or other similar rights in any country, in each case whether or not registered, and any applications for registration of any of the foregoing, and all rights to apply to register any of the foregoing.

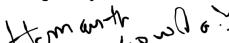
“Inventions” means any and all inventions, solutions, specifications, recommendations, software, user manuals, documentation, reports, designs, drawings, photographs, maps, data or other information, artistic, literary or musical works, films, sound recordings, documents or materials of any nature and in any medium which are made, created or generated by You in the course of Your employment, whether or not made during normal office hours and whether or not made using Royal Cyber facilities or equipment.

2. Confidential Information

- 2.1 You shall not, without the prior written consent of Royal Cyber use Confidential Information for any purpose other than Your employment or disclose Confidential Information to any third party.
- 2.2 The obligations set out in Clause 2.1 shall cease to apply to information after such time as it is or becomes public knowledge through no fault of your own or is or becomes lawfully known to You from another legitimate source without restrictions on use or disclosure. It shall be for You to establish the application of any such exception with adequate proof. A piece of information shall not be deemed within the public domain merely because it can be broken down into constituent parts each of which can be shown separately to be in the public domain.
- 2.3 Notwithstanding the foregoing, you may disclose Confidential Information as may strictly be required by law or ordered by a Court of competent jurisdiction, provided that you shall if legally possible immediately notify Royal Cyber of the requirement to disclose and allow Royal Cyber the opportunity to seek a protective order or other remedies.

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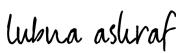
- 2.4 You shall not at any time remove any Confidential Information in hard copy, soft copy or electronic form from the possession of Royal Cyber except so far as may strictly be necessary for the performance of Your employment and with the permission of Your supervisor. On the termination of Your employment, or earlier if requested by Royal Cyber, You shall immediately return to Royal Cyber (or at Royal Cyber's option destroy) all Confidential Information in hard copy, soft copy or electronic form in Your possession or control and shall not retain copies of such material, communication or documents.
- 2.5 You shall have no license under any Intellectual Property Rights of Royal Cyber save as necessary for the performance of your employment.
- 2.6 It is Royal Cyber's policy to encourage each Intern to take all reasonable opportunities to address or read papers to meetings of learned societies and to cause original papers to be published in the proceedings of such societies or organizations, or in trade and technical journals, provided always that on each occasion on which such an address is to be delivered or paper read or published, the content or text is first approved by Royal Cyber in its absolute discretion if such content or text contains or may contain Confidential Information.
- 2.7 You agree not to compete with Royal Cyber directly or indirectly (including as an officer, Intern, director, subcontractor or agent) from the time of Your acceptance of this offer, during the term of Your employment and for a period of 24 months thereafter.

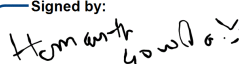
3. Inventions

- 3.1 If at any time during the performance of Your employment you make, or contribute to the making of, any Invention, You shall forthwith supply Royal Cyber with full particulars of such Invention. You shall not disclose or give details of any Invention to any person whatsoever except Royal Cyber or its representatives.
- 3.2 Royal Cyber shall own and have the exclusive right to exploit Inventions, and You hereby assign to Royal Cyber by way of future assignment and with full title guarantee all Intellectual Property Rights in Inventions. Royal Cyber shall have the exclusive right to file, at its expense, applications for patents or other Intellectual Property Rights for Inventions in or for whatever country or countries as Royal Cyber may deem fit and to exploit any patent or other right issuing from any such application as it deems fit.
- 3.3 Without prejudice to the assignment in Clause 3.2, You shall if necessary and at the expense of Royal Cyber, provide Royal Cyber or Royal Cyber's nominee with all such documents and information and render such assistance and execute all such instruments and otherwise do all such acts as Royal Cyber may Reasonable require to transfer Intellectual Property Rights in Inventions to Royal Cyber or its nominee and to enable Royal Cyber to secure protection for those rights.

4. Background Information and Rights

- 4.1 You shall not use or incorporate in any work or materials You produce in the course of your employment, or disclose to Royal Cyber or its representatives, any information or other items in respect of which You are under obligations of restricted use or disclosure to a third party, save where You have obtained (at no cost to Royal Cyber) prior to such use, a license for Royal Cyber on the terms set out in Clause 4.2 in respect of such information or items.
- 4.2 To the extent that You are entitled to grant such a license, You hereby grant to Royal Cyber a non-exclusive, royalty-free, irrevocable and worldwide license for Royal Cyber with sub-licensing rights under Background

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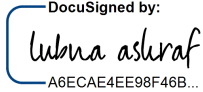
Signed by:

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Rights as necessary to use, copy, modify and distribute freely any Inventions or materials delivered by You to Royal Cyber in the performance of Your employment.

5. Governing Law and Jurisdiction

This agreement shall be governed by and construed in accordance with the laws of India and the courts of competent jurisdiction at Bangalore shall have the exclusive jurisdiction to settle any dispute arising from or connected with this Agreement between parties hereto.

For Royal Cyber Pvt Ltd

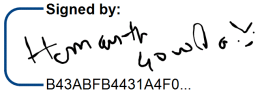
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Lubna Ashraf

Global HR Head

ACCEPTANCE OF APPOINTMENT TERMS & CONDITIONS

I, Hemanth Gowda, agree that I have read, understood, and I voluntarily accept and sign this as per the terms & conditions stated above.

Signature: 
B43ABFB4431A4F0...

Name: Hemanth Gowda

Date: 2/7/2025 | 9:54 AM PST

ACKNOWLEDGEMENT

This internship is the result of continuous guidance, support, and encouragement from several individuals, to whom I am deeply grateful. I take this opportunity to express my sincere appreciation to everyone who contributed to the successful completion of this internship.

I would like to express my heartfelt thanks to **Dr. B G Prasad**, Principal, Dayananda Sagar College of Engineering, for providing the necessary facilities and support to undertake this internship.

My sincere gratitude goes to **Dr. Vindhya P Malagi**, Head of the Department – Artificial Intelligence & Machine Learning, for providing me the opportunity to pursue this internship and for her valuable guidance and encouragement throughout.

I am thankful to **Prof. Kavya D N**, Internship Coordinator and Assistant Professor – AI & ML, for her continuous support, timely suggestions, and motivation during the internship period.

I would also like to extend my thanks to **Prof. Deepshree Buchade**, internal guide, for her consistent guidance, timely feedback, and unwavering support throughout the internship. Her mentorship helped me stay aligned with both academic and industry expectations.

I would like to thank **Guru**, Senior ServiceNow Developer, Royal Cyber, for the valuable industry insights and mentorship provided during my role as an intern in ServiceNow Developer.

My thanks also go to all the faculty members and non-teaching staff of the Department of AI & ML for their cooperation and assistance during the internship period.

Hemanth Gowda V

[1DS21AI022]

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CHAPTER 1

INTRODUCTION

In today's dynamic enterprise IT landscape, platform efficiency, user experience, and process automation have become pivotal for long-term organizational growth. As businesses increasingly adopt digital transformation initiatives, platforms like ServiceNow play a crucial role in streamlining operations, improving service delivery, and enhancing internal stakeholder satisfaction. This project centers around analyzing platform performance and user interaction metrics such as request volume by category, fulfillment timelines, SLA compliance, and incident resolution efficiency—key indicators that reflect the effectiveness of enterprise service management and developer-led enhancements.

Through detailed analysis of user engagement and transaction patterns across modules like ITSM, HRSD, and CSM, the project identifies trends in platform adoption, common pain points, and potential areas for workflow optimization. These insights are essential for ServiceNow developers and administrators, enabling them to fine-tune system behavior, improve UX/UI elements, and tailor service catalogs for high-impact delivery. The project leverages cohort analysis to observe system usage trends based on onboarding cohorts or feature rollout periods—providing a structured way to monitor the long-term success of new implementations and developer customizations.

Additionally, the initiative includes the development of a predictive model for user disengagement or request drop-off, built using historical platform data and user interaction logs. This model incorporates features like request frequency, resolution delays, and user feedback scores, and applies classification algorithms to anticipate at-risk users or departments. These insights empower proactive improvements in platform configurations, user training, and support strategies—areas directly influenced by ServiceNow developers.

Ultimately, the objective of the project is to harness ServiceNow analytics and development insights to drive continuous platform improvement. By combining descriptive metrics with predictive modeling, the work supports data-driven decisions, boosts system adoption, and enhances overall IT service efficiency—solidifying the role of ServiceNow developers in the success of enterprise operations.

CHAPTER 2

ABOUT THE COMPANY



1.1 Introduction to ServiceNow

ServiceNow is a leading cloud-based platform that offers digital workflow solutions to streamline and automate business processes across various departments, such as IT, Human Resources, Customer Service, Security Operations, and more. Originally introduced as an IT Service Management (ITSM) tool, ServiceNow has rapidly evolved into a comprehensive enterprise platform capable of driving digital transformation and operational efficiency throughout organizations.

At its core, ServiceNow provides a single system of record for all IT processes within an organization, enabling seamless integration, automation, and visibility across the entire IT landscape. It supports ITIL-based service management, ensuring best practices in handling incidents, problems, changes, requests, and configurations. This ensures that IT departments can deliver services more efficiently while maintaining compliance and governance standards.

Beyond IT, ServiceNow extends its capabilities through purpose-built workflows and modules for various business functions. For example, HR Service Delivery (HRSD) enables HR teams to deliver services through self-service portals and case management, reducing manual intervention and improving employee experience. Similarly, Customer Service Management (CSM) helps organizations resolve customer issues proactively by integrating support with other business functions, such as operations and field services.

In terms of innovation, ServiceNow integrates emerging technologies such as Artificial Intelligence (AI), Machine Learning (ML), and Predictive Analytics to enhance decision-making, automate routine tasks, and deliver personalized user experiences. For instance, Virtual Agents powered by AI handle common user queries, while Predictive Intelligence helps categorize and route tickets efficiently. Overall, ServiceNow acts as a digital backbone for modern enterprises, fostering collaboration, transparency, and agility.

1.2 Introduction to Royal Cyber



Royal Cyber Inc. is a global IT consulting and digital transformation company that specializes in providing cutting-edge technology solutions to a wide array of industries. Founded in 2002 and headquartered in Naperville, Illinois, USA, Royal Cyber has established itself as a trusted partner for businesses seeking innovation, efficiency, and scalable growth through digital modernization.

Over the years, Royal Cyber has expanded its presence across the globe with offices in North America, the Middle East, Asia Pacific, and Australia. The company serves a diverse client base, ranging from Fortune 500 enterprises to mid-sized businesses, by delivering tailored solutions that address complex technological challenges. Their global delivery model and strong emphasis on client collaboration enable them to deliver consistent and high-quality outcomes.

1.2.1 Core Areas of Expertise:

Royal Cyber offers services across several technological domains, making it a one-stop solution for businesses undergoing digital transformation. Their core competencies include:

- Digital Commerce – Providing end-to-end solutions for e-commerce platforms using tools like SAP Commerce Cloud, Salesforce Commerce Cloud, and Adobe Commerce (Magento).
- Cloud Services – Assisting organizations with cloud migration, cloud-native development, and managed services across platforms like AWS, Azure, and Google Cloud.
- DevOps & Automation – Enabling CI/CD, containerization, and infrastructure automation to help businesses achieve agility and faster time-to-market.
- Data & AI – Empowering businesses with intelligent analytics, data engineering, and AI/ML solutions for better decision-making.

1.2.2 Commitment to Innovation and Quality:

What sets Royal Cyber apart is its strong focus on innovation and customer-centricity. The company invests heavily in research and development, offering accelerator tools, pre-built frameworks, and proof-of-concept models that reduce time-to-market and minimize risk. Their teams are composed of certified professionals who stay ahead of emerging technologies to deliver future-ready solutions.

Moreover, Royal Cyber is known for its collaborative approach—working closely with clients to understand their goals, challenges, and organizational culture. This results in solutions that are not only technologically sound but also aligned with business objectives.

1.2.3 Global Presence and Industry Reach:

Royal Cyber serves a diverse clientele spanning a wide spectrum of industries, including retail, manufacturing, finance, logistics, healthcare, telecommunications, education, and government services. Their deep understanding of domain-specific challenges allows them to design highly customized and context-aware digital solutions.

Their global delivery model ensures round-the-clock support and project execution capabilities. With development and innovation hubs in India, Pakistan, Saudi Arabia, UAE, Kenya, and Australia, Royal Cyber is uniquely positioned to offer localized support with global quality standards.

Culture, Values, and Workforce:

Royal Cyber is known for its diverse, inclusive, and innovation-driven culture. The company employs over 1,500 professionals worldwide, including solution architects, cloud engineers, software developers, UX designers, and data scientists. A strong emphasis is placed on certification, training, and continuous learning, ensuring that its consultants are well-versed in the latest tools and methodologies.

Its core values include Customer Obsession, Agility, Collaboration, Continuous Innovation, and Integrity. These values are embedded in every project and client interaction, ensuring lasting relationships and transformative outcomes.

1.3 Founder



Mustafa Peshawarwala is the **founder** and current **CEO** of Royal Cyber Inc., a company he established in **2002** with a vision to create a leading IT consulting and digital transformation firm that would help global enterprises adapt to fast-evolving technologies.

With a strong academic foundation in Computer Science from the Illinois Institute of Technology, Mustafa brings a blend of deep technical knowledge and sharp business acumen. Over the past two decades, he has led Royal Cyber from a small start-up to a global digital powerhouse with a presence in over 10 countries including the USA, Canada, Saudi Arabia, UAE, India, Pakistan, Kenya, and Australia.

Under his leadership, Royal Cyber has:

- Partnered with Fortune 500 clients.
- Earned multiple accolades such as the Best Places to Work, CRN Tech Innovator, and Digital SaaS Partner of the Year.
- Built strong alliances with technology giants like IBM, ServiceNow, Salesforce, AWS, Microsoft Azure, and MuleSoft.

Cultivated a workplace culture centered on innovation, integrity, and inclusiveness. Mustafa is widely regarded for his forward-thinking mindset, especially in areas like AI, DevOps, Cloud Computing, and Enterprise Integration, and is often invited to industry events to speak on leadership in digital transformation

CHAPTER 3

SCOPE OF THE WORK

1. Digital Transformation Services

Royal Cyber’s core expertise lies in enabling businesses to evolve digitally by leveraging next-generation technologies. Their digital transformation services focus on modernizing existing IT landscapes, automating workflows, and enabling intelligent business operations. These services include cloud adoption, artificial intelligence integration, data analytics, and workflow automation—delivered through a consultative and agile approach. The company ensures that each solution aligns with the client’s strategic vision and improves time-to-market, scalability, and operational efficiency.

2. Application Development and Modernization

One of Royal Cyber’s foundational capabilities is custom application development tailored to client-specific needs. Whether it’s a web-based enterprise application or a scalable mobile solution, Royal Cyber employs modern frameworks such as React, Angular, Node.js, and Spring Boot to design, build, and maintain high-performing applications. Additionally, legacy systems are transformed into modern microservices architectures that support better integration, security, and performance. The adoption of DevOps practices ensures rapid development cycles and consistent delivery.

3. ServiceNow Consulting and Platform Implementation

As a certified elite partner of ServiceNow, Royal Cyber delivers end-to-end services across the platform. The scope includes:

- **IT Service Management (ITSM)** for streamlining incident, problem, and change management.
- **IT Operations Management (ITOM)** to enhance infrastructure visibility and reduce downtime.
- **HR Service Delivery (HRSD)** for automating employee-related services and onboarding workflows.
- **Customer Service Management (CSM)** to improve client engagement through self-service portals and omnichannel support.

- **Custom Application Development** using ServiceNow's App Engine for tailored business processes.

These implementations are supported by extensive integration capabilities and post-deployment support.

4. Industry-Specific Solutions

Royal Cyber customizes its services to suit the regulatory and operational needs of various industries:

- **Retail & eCommerce:** End-to-end commerce platforms, order management, and customer analytics.
- **Healthcare:** HIPAA-compliant applications, patient portals, and appointment systems.
- **Banking & Finance:** Secure transactions, risk management, and financial reporting tools.
- **Logistics & Supply Chain:** Real-time tracking, inventory systems, and ERP integrations.
- **Education and Government:** e-Governance portals, student information systems, and digital classrooms.

5. Cloud Solutions and Infrastructure Services

Royal Cyber offers cloud consulting, migration, and infrastructure management services across major cloud platforms including AWS, Microsoft Azure, and Google Cloud. The scope includes cloud readiness assessments, hybrid cloud architecture design, and infrastructure automation using tools like Terraform and Ansible. Containerization strategies using Docker and Kubernetes further enhance scalability and deployment speed. This focus enables clients to reduce costs, increase agility, and ensure robust performance for mission-critical applications.

CHAPTER 4

LEARNING OBJECTIVES

1. Understand the Fundamentals of ServiceNow and Its Ecosystem

Participants will gain a solid understanding of **ServiceNow**—one of the most widely used cloud platforms for IT Service Management (ITSM), IT Operations Management (ITOM), HR Service Delivery (HRSD), and more. Learners will:

- Explore ServiceNow architecture, platform capabilities, and workflow automation features.
- Understand the core modules: Incident, Change, Problem, Knowledge, and Asset Management.
- Learn how ServiceNow integrates with third-party systems and enhances enterprise productivity.

2. Gain Practical Experience in ServiceNow Application Development

A key objective is to enable participants to **design, build, and configure applications within the ServiceNow platform**. This includes:

- Creating custom tables, forms, and UI elements.
- Implementing client-side and server-side scripting (Business Rules, Client Scripts, UI Policies).
- Configuring Access Control Lists (ACLs) to manage data security and visibility.
- Developing workflows using Flow Designer and Script Includes.

3. Learn the Principles of Digital Transformation and Cloud Technologies

Learners will delve into the **strategic and technical aspects of digital transformation**, including:

- Understanding how businesses evolve through digital tools.
- Identifying challenges in legacy systems and how to modernize them.
- Exploring cloud service models (IaaS, PaaS, SaaS) and platforms such as AWS, Azure, and Google Cloud.

- Examining real-world case studies of successful transformation initiatives led by Royal Cyber.

4. Build Skills in Agile Development and DevOps Practices

To prepare for dynamic and modern software development environments, the learning path emphasizes:

- The principles of Agile methodologies and Scrum framework.
- How to work in sprints, maintain backlogs, and conduct reviews.
- Basics of DevOps, including CI/CD pipelines, version control (Git), automated testing, and deployment strategies.

5. Explore Integration Techniques and Middleware Solutions

Integration is critical in digital ecosystems. This objective focuses on:

- Understanding middleware platforms such as **MuleSoft, IBM Integration Bus, and Apache Kafka.**
- Learning how to design and implement REST and SOAP APIs.
- Creating data pipelines and event-driven architectures to connect multiple enterprise systems seamlessly.

6. Develop Customer-Centric Thinking through CRM and CSM Solutions

Participants will study how technologies are used to improve customer interactions and support services. Objectives include:

- Learning about Customer Relationship Management (CRM) and Customer Service Management (CSM) platforms.
- Configuring portals and communication channels to enhance customer experience.
- Understanding case management, automated responses, and service-level agreements (SLAs).

CHAPTER 5

CHALLENGES AND SOLUTIONS

1. Adapting to a Complex Platform like ServiceNow

ServiceNow is a feature-rich and expansive cloud platform with multiple modules and configurations. As a newcomer, understanding its ecosystem—ranging from workflows, ACLs, and scripting to UI policies and integration—proved initially overwhelming. A structured learning path was followed, starting with ServiceNow fundamentals and then diving into specific modules such as ITSM and HRSD. Hands-on practice through lab environments helped in understanding configurations and scripting techniques. Regular mentorship and internal documentation provided by Royal Cyber experts played a crucial role in bridging knowledge gaps.

2. Understanding Real-World Business Requirements

Unlike academic scenarios, real-world business requirements are often ambiguous, evolving, and domain-specific. Translating client needs into technical specifications required critical thinking and contextual understanding. Participated in requirement gathering meetings and stakeholder discussions to gain clarity. Learned to use user stories and business process flow diagrams to structure requirements effectively. Engaged in iterative reviews and feedback loops to ensure that development aligned closely with client expectations.

3. Platform-Specific Scripting and Configuration Issues

Working with ServiceNow scripting (e.g., **Business Rules**, **Client Scripts**, **Script Includes**) introduced challenges related to syntax, data handling, and debugging errors that were not immediately obvious. Utilized ServiceNow's scripting documentation and developer forums for issue resolution. Followed best practices such as using GlideRecord efficiently and handling server-side vs. client-side logic properly. Applied breakpoints and `gs.log()` functions for debugging, making troubleshooting more structured and effective.

4. Managing Time Across Multiple Tasks and Deadlines

Balancing between learning, development tasks, documentation, and meetings created pressure and sometimes led to overlapping responsibilities. Adopted Agile practices such as creating sprint plans, daily to-do lists, and prioritizing tasks using tools like **Jira** and **Trello**. Maintained a personal Kanban board to track progress and blockers. Practiced effective communication to clarify workloads and negotiate task adjustments when needed.

5. Integration with External Systems

In scenarios where ServiceNow needed to be integrated with third-party tools (like email systems, databases, or CRM platforms), configuration and API understanding became complex. Explored **IntegrationHub** and REST APIs within ServiceNow to understand outbound/inbound integration flows. Collaborated with system architects and senior developers to learn the structure of integration patterns. Tested API endpoints using tools like **Postman** before implementing them in real environments.

5. Deployment and Testing Issues

While promoting updates from the development environment to testing or production, some configuration inconsistencies or missing dependencies led to deployment errors. Learned to use **Update Sets** effectively and documented all dependencies clearly. Established a checklist before any environment migration. Performed thorough UAT (User Acceptance Testing) and regression tests to ensure a smooth transition.

6. Access and Permission Constraints (ACL Configuration)

Implementing secure data access using Access Control Lists (ACLs) was initially tricky, often leading to unintended restrictions or exposure of data. Gained a clear understanding of ServiceNow's ACL evaluation order (Record → Table → Field). Used Security_Debug mode to track why access was denied or granted. Created role-based access structures aligned with best practices and business needs.

CHAPTER 6

ACHIEVEMENTS AND CONTRIBUTION

1. Successfully Developed a Custom ServiceNow Scoped Application

One of the most significant contributions was the successful development of a **custom ServiceNow Scoped Application**—"NeedIT"—tailored to address specific organizational needs.

Key Achievements: Created a **custom table** with defined fields and relationships. Configured form layouts to enhance usability and improve data entry workflows. Applied **UI Policies** to control field visibility and behavior dynamically based on record states. Implemented custom buttons (e.g., "Close", "Reopen") with visibility rules and scripted actions. Ensured proper access control through **ACL configuration** to maintain data security and role-based access.

2. Automation of Workflows Using Flow Designer

Another important achievement was the automation of repetitive tasks and business processes using ServiceNow's **Flow Designer**.

Key Contributions: Designed and implemented flows to trigger actions based on specific events such as form submission or status change. Integrated decision logic and email notifications within flows to improve responsiveness and reduce manual overhead. Documented flows for easy maintenance and handover to future team members.

3. Contribution to Team Collaboration and Agile Practices

Beyond technical tasks, active participation in team processes and collaboration added value to the overall project environment.

Specific Contributions: Participated in **daily stand-ups**, sprint planning, and review meetings. Provided regular status updates and raised blockers in a timely manner. Contributed to project documentation, user manuals, and configuration records for better knowledge sharing.

4. Strengthened Platform Knowledge Through Hands-On Learning

A strong personal achievement was the accelerated learning curve in understanding complex platform capabilities and ServiceNow architecture.

Highlights: Gained proficiency in client-side (Client Scripts, UI Policies) and server-side scripting (Business Rules, Script Includes). Understood data modeling, relationships, and

platform-specific terminology (e.g., GlideRecord, GlideForm). Applied debugging techniques and utilized developer tools to resolve scripting and configuration issues.

5. Shadowing in a Real-Time Project – Bombardier

One of the most insightful experiences during the internship was the opportunity to **shadow a live ServiceNow project for Bombardier**, a global leader in aviation and rail transportation solutions. This experience offered a first-hand view into how enterprise IT solutions are conceptualized, executed, and maintained in high-stakes environments.

- **Observed and Assisted in Daily SME Activities:** Regularly joined SME meetings, sprint discussions, and task updates to stay aligned with project developments. Closely followed how incidents, requests, and customizations were handled in a real-time ServiceNow environment.
- **Clarified Project-Specific Queries:** Took the initiative to document doubts and technical blockers, and worked alongside the SME to resolve them through direct interaction or independent research. This significantly improved understanding of platform-specific implementations used in Bombardier.
- **Knowledge Backup and Support:** Acted as a reliable backup for the SME by understanding the workflows, configuration logic, and data flow within the project. Was prepared to take over or assist in tasks when the SME was unavailable, ensuring continuity of progress.
- **Maintained Documentation:** Helped update and organize key project artifacts including configuration logs, development checklists, and client-specific customization notes for ease of future reference and handover.

Enabled a comprehensive understanding of large-scale project dynamics, specifically within a global client scenario. Built confidence in handling platform configurations independently. Improved communication and reporting skills by emulating SME practices. Strengthened preparedness to step into support or enhancement roles with minimal onboarding.

CHAPTER 7

REFLECTION AND ANALYSIS

The professional engagement at **Royal Cyber Inc.** represented a comprehensive and immersive experience, combining technical learning, real-time project exposure, and business acumen. This opportunity facilitated not only the enhancement of my technical capabilities but also fostered a deep understanding of the operational standards, methodologies, and competencies required within the IT consulting domain. Reflecting on this journey reveals key insights into my professional development, adaptability, and potential contributions to enterprise technology environments.

1. Technical Enrichment and Platform Expertise

A major focus of the internship was hands-on engagement with the **ServiceNow platform**, an industry-leading solution for digital workflow automation. The development of a custom scoped application—encompassing table design, ACL configuration, client and server-side scripting, and UI customization—significantly strengthened my ability to architect and implement process-driven solutions aligned with organizational needs.

In parallel, shadowing an SMEs on the **Bombardier project** provided real-time exposure to advanced ServiceNow implementations in a client-facing setting. This allowed for the assimilation of best practices, from incident resolution and module configuration to stakeholder communication. Serving as a knowledge backup to the SME further instilled confidence in navigating platform intricacies and resolving technical queries independently.

2. Exposure to Agile Methodologies and Team Collaboration

Participating in Agile routines such as daily stand-ups, sprint reviews, and backlog refinement meetings offered a structured framework for task prioritization, team coordination, and iterative delivery. This exposure enhanced my ability to function in fast-paced environments, adapt to changing requirements, and communicate progress effectively across cross-functional teams.

Collaborating with professionals across development, QA, and project management not only deepened my understanding of cross-role interactions but also highlighted the value of transparency, accountability, and structured reporting in a professional services setting.

3. Strategic Understanding of Business and IT Alignment

Through involvement in sprint planning sessions, workflow automation tasks, and documentation processes, I gained a clear perspective on how IT solutions must be tailored to meet specific business objectives. Understanding the nuances of modules like **ITSM**, **ITOM**, **HRSD**, and **CSM**, and how they support organizational service delivery, was critical to appreciating the broader context in which technical systems operate.

Moreover, the internship underscored the importance of aligning technical solutions with end-user experience, governance standards, and continuous service improvement models—key principles that govern enterprise IT operations.

4. Problem Solving and Adaptive Thinking

Throughout the course of the engagement, I encountered several real-world technical and functional challenges—from debugging business rules to configuring access control models in accordance with organizational security protocols. These scenarios demanded a high level of analytical thinking, platform knowledge, and resourcefulness. Leveraging documentation, peer support, and self-directed research became instrumental in delivering efficient solutions.

This iterative problem-solving approach enhanced my technical fluency and demonstrated the value of a growth mindset in overcoming complex project scenarios.

In summary, the experience at Royal Cyber Inc. was profoundly impactful. It provided a dynamic blend of technical depth, strategic insight, and professional maturity. This reflection affirms the significant competencies gained—not only in ServiceNow development but also in agile project execution, stakeholder engagement, and enterprise readiness. The insights and skills acquired during this tenure serve as a strong foundation for continued growth and contribution in the IT consulting and digital services landscape.

CHAPTER 8

RECOMMENDATIONS

A more **structured onboarding process** at the beginning of the internship can significantly enhance early-stage productivity. While initial orientation was helpful, a detailed walkthrough of existing projects, team structures, and ongoing deliverables could help new interns integrate faster.

Suggested Improvements:

- Provide a dedicated onboarding document or knowledge base with essential platform links, project context, coding standards, and access instructions.
- Schedule formal knowledge-sharing sessions with mentors and SMEs in the initial weeks.
- Assign guided walkthroughs of existing applications to understand real implementations.

While working on tasks and shadowing on the Bombardier project, it became clear that **well-maintained documentation** was vital in understanding past decisions, configurations, and workflows. Encourage teams to maintain up-to-date technical documentation for all modules and scripts. Include inline comments and rationale in script includes, business rules, and UI policies. Introduce documentation templates for developers to follow, ensuring consistency across the platform.

The shadowing experience under the SME proved to be one of the most enriching aspects of the internship. Expanding such opportunities can bridge the gap between theoretical learning and real-time project execution. Formally integrate **shadowing programs** as part of the internship structure, assigning interns to specific senior professionals for a defined period.

Facilitate cross-training sessions where interns can get exposed to different ServiceNow modules like **ITOM, HRSD, SecOps**, etc. Allow interns to assist in internal testing or UAT to experience the full SDLC. Understanding client requirements and project delivery expectations is a vital part of any consulting role. Interns can benefit greatly from limited exposure to these interactions.

Allow interns to attend selected internal project review meetings or client demo rehearsals. Encourage interns to document mock client requirements and propose solutions as a learning exercise. Include communication-focused training to enhance email etiquette, task updates, and reporting.

CHAPTER 9

CONCLUSION

The internship experience has been both professionally enriching and personally transformative. It provided a rare opportunity to engage with real-world projects, cutting-edge technology, and a dynamic team of industry experts. Working primarily with the **ServiceNow platform**, I was able to translate theoretical knowledge into practical implementations, enhancing both my technical capabilities and professional competencies.

Throughout the course of this internship, I was entrusted with meaningful responsibilities, including the development of a **scoped application**, involvement in key platform configurations, participation in sprint routines, and the privilege of **shadowing an SME** on the Bombardier project. These experiences offered invaluable insights into the lifecycle of enterprise application development—from requirement gathering and design, through development and deployment, to maintenance and client engagement.

The internship also fostered a deep understanding of how IT services align with business goals through solutions like **ITSM, ITOM, HRSD, and CSM**. Exposure to these modules not only enhanced my technical foundation but also broadened my perspective on service delivery, user experience, and operational efficiency in enterprise environments. Equally important was the development of soft skills such as time management, professional communication, adaptability, and collaborative problem-solving. Regular feedback, knowledge-sharing sessions, and agile project involvement helped reinforce these aspects, preparing me for the demands of a modern IT consulting role.

Moreover, the organization's inclusive and mentorship-driven culture played a crucial role in accelerating my learning curve. The support from mentors, peers, and project leads cultivated an environment of continuous improvement and inspired me to take initiative, seek clarity, and contribute meaningfully. In reflection, this internship was not merely a training period—it was a pivotal chapter in my professional journey. It has provided a clear direction for my career aspirations, built a strong technical foundation, and instilled the confidence required to thrive in a global, fast-paced digital services industry. As I move forward, I carry with me the skills, knowledge, and values acquired during this experience, fully prepared to take on new challenges and contribute with purpose and professionalism.